



Hands-on Principal Architect · 19+ yrs · Python + Java · Data Engineering + MLOps · Event-Sourced Pipelines · Parquet + DuckDB · Agentic AI + RAG · Neptune + GNN · Distributed Systems at Scale

SUMMARY

Hands-on principal architect, 19+ years, with data engineering running the whole length of it — master data management and field-level lineage across North America in 2012, entity resolution over 25 supplier catalogues on a platform serving 2–3 million requests daily in 2016, and a PostgreSQL to Amazon Aurora estate through 2025. Then the data and artificial intelligence (AI) platform for a chronic-care product on Amazon Web Services (AWS): an append-only DynamoDB event spine holding 9–10 years of patient history alongside live traffic, deterministic Parquet snapshots read by DuckDB, and ten production services in Python on FastAPI — retrieval-augmented generation over dense vectors, Amazon SageMaker retraining gated on drift detection and promotion tests. Java and Spring Boot across roughly 11 years, through Spring Boot 2.x on Java 8/17 to February 2025. Brought infrastructure cost down 25–30% and deploy time from hours to under 10 minutes.

Key Skills

Technology Leadership & Strategy

- Enterprise Platform Architecture (End-to-End Ownership, Long-Horizon Evolution)
- Data Platform & Analytics Leadership
- Technology Strategy & Company-Wide Standards Setting
- Decision-Intelligence Platforms (Event-Sourced + Graph + AI)
- Engineering Team Building & Organisational Scaling
- Cross-Functional Leadership & Stakeholder Alignment

Engineering Excellence

- Python Backend Engineering (FastAPI, Django, LangChain, LangGraph, Pydantic)
- Data Engineering at Scale (Event Sourcing, Command Query Responsibility Segregation (CQRS), Deterministic Parquet Snapshot Exporters, DuckDB)
- Pipeline Correctness (Idempotent Backfill, Content-Addressed Event Identifiers, Schema Upconverters, Quarantine-First Loads, Golden Replay)
- Streaming & Distributed Ingestion (Amazon Kinesis Enhanced Fan-Out, Simple Queue Service, Lambda, OpenSearch)
- Machine Learning Operations (MLOps) (Retraining Pipelines, Promotion Gates, Drift Detection, Model Versioning, Batch Transform, Self-Hosted LangSmith)
- Deep Learning in Production (Amazon SageMaker Endpoints, Sequence Models, Self-Hosted Embeddings, graph neural network (GNN) on Amazon Neptune)
- Retrieval-Augmented Generation (RAG) & Agentic AI (Filtered Vector Search, Elasticsearch dense_vector, LangGraph Workflows)
- Leakage-Safe Machine Learning (As-Of Featurisation, Point-in-Time Correctness, Out-of-Time Back-Tests, Outcome-Maturity Windows)
- Large Language Model (LLM) Productionisation (Guardrails, Field-Level Claim Grammar, Replay Determinism)
- Statistical Rigor in Production AI (Propensity Matching, Inverse Probability Weighting, Standardised Mean Difference Balance Gates, Bootstrap Confidence Intervals)
- Java + Spring Boot Microservices (Java 8/17, Spring Boot 2.x, JSP — 11 years across three eras)
- Distributed Systems & Polyglot Persistence (DynamoDB, PostgreSQL, Amazon Aurora, OpenSearch, Redis, Amazon Neptune + Gremlin)
- Multi-Tenant Software-as-a-Service (SaaS) Architecture (Shared-Schema, Per-Tenant Isolation, Regulated Carve-Outs)
- Cloud Infrastructure & Platform Engineering (AWS, Kubernetes, Terraform, Elastic Container Service (ECS) Fargate)
- Shared Platform Application Programming Interfaces (APIs) (Contracts, Versioning, Backward Compatibility)
- DevOps & Continuous Integration / Continuous Delivery (CI/CD) (Jenkins to GitHub Actions)
- Observability & Reliability Engineering (Datadog, Amplitude, Elasticsearch, Logstash and Kibana (ELK))

Business & Delivery

- Agile / Scrum Delivery Management
- Security, Compliance & Data Privacy (Least-Privilege, Per-Tenant Isolation, Protected Health Information Boundaries, ISO 27001-Certified Environment)
- Technical Mentoring, Design Review & Architectural Decision-Making
- Performance Optimization & Cost Management
- Enterprise Onboarding & Consolidation at Multi-Hundred-Instance Scale
- Distributed United States–India Delivery Teams

PROFESSIONAL EXPERIENCE

Principal Software Architect Mar '25 - Apr '26

Voltuswave Technologies India Pvt Ltd Hyderabad

- Instituted event sourcing on DynamoDB single-table design with a deterministic Parquet snapshot exporter read by DuckDB, targeting P95 rebuild within 5 minutes on correction
- Hydrated nine years of history through the same append-only spine as live traffic — schema upconverters, quarantine on unmappable rows, four-field event-time provenance, re-runs byte-identical
- Sealed multi-tenant isolation across 10 production pipelines — a tenant-scoped traversal-source wrapper, Parquet partitioning, tenant filters at the columnar scan and per-row runtime assertions

- Engineered a Python and FastAPI RAG pipeline on filtered Elasticsearch dense_vector search across 80+ chronic conditions, keeping Amazon Neptune off the serving path - Distilled patient event histories into a relapse early-warning model on Amazon SageMaker, gated through a three-model ladder — logistic, PrefixSpan, Transformer — on out-of-time back-tests - Triggered weekly retraining on SageMaker Pipelines from tokenizer and reliability drift, scoring nightly by batch transform over pre-materialised 30-day windows, endpoint swapped behind promotion gates - Provisioned Kubernetes and Terraform as a company-wide standard on ECS Fargate at 70% Spot — deploys under 10 minutes, infrastructure cost down 25–30%	Dec '19 - Feb '25
Lead SDE/Sr. Staff Engineer / Staff Software Engineer	
Deque Software	Hyderabad
- Implemented multi-tenant isolation across approximately 300 enterprise instances, adding an organisation ID to every table and resolving tenant from the JSON Web Token (JWT) - Steered 70–80% of customers onto shared axe Monitor instances on Spring Boot 2.x and Java 8/17, one codebase also serving regulated banks on dedicated databases - Stewarded a zero-downtime upgrade from PostgreSQL 9.6 to Aurora PostgreSQL 13.x across every axe Auditor instance, cutting backups 3–4 hours daily and deploy errors 40% - Orchestrated scheduled website-discovery and axe-core scanning pipelines, diffing consecutive runs into three states — new, fixed and regressed uniform resource locators (URLs) - Championed decomposition of a monolith into 10–15 services, splitting on independent scaling and database ownership rather than on team boundaries - Exposed shared platform APIs to 20+ enterprise clients — Jira integration, API versioning, customer-specific frontends on one Spring Boot backend, backward compatibility - Modernised CI/CD from Jenkins to GitHub Actions, achieving ISO 27001 and 50% faster release cycles	
Software Engineer III	Aug '18 - Jul '19
Rocket Software	Bangalore
- Anchored decomposition of IBM Db2 Configuration Manager into Java and Spring Boot microservices, defining boundaries across z/OS and Linux, UNIX and Windows platforms - Coached a 5-engineer team through delivery planning, code review and performance appraisals while remaining hands-on as an individual contributor - Surfaced defect hot-spots through REST interfaces and service isolation, reducing reported bugs over 50% - Partnered with product to extend platform support to Windows and Linux — Ubuntu, CentOS and Red Hat Enterprise Linux (RHEL) — raising sales over 20%	
Co-Founder and VP of Technology	Mar '17 - Apr '18
Voltuswave Technologies India Pvt Ltd	Hyderabad
- Co-founded VoltusWave and launched its first application Platform-as-a-Service (aPaaS) product for web and Android, winning 8 early enterprise clients - Mobilized an engineering organisation from 0 to 35 in 13 months, mentoring freshers and senior engineers while establishing delivery process - Helmed development, deployment and operations on Amazon Elastic Container Service, setting the architecture roadmap every subsequent product inherited	
Senior Software Engineer	Mar '16 - Mar '17
Teletext India Private Limited, Hyderabad	Hyderabad
- Codified a scatter-gather integration across 25 hotel suppliers on a MuleSoft enterprise service bus, succeeding where a prior .NET 2.0 upgrade had failed - Cut operational cost 46% by retiring a third-party pricing product for an in-house ELK cache with a 20-minute time-to-live - Reconciled supplier hotel identifiers onto GIATA MultiCodes, automating entity resolution across 25 heterogeneous supplier catalogues on AWS Lambda, growing revenue 8% month over month - Lifted hotel booking conversions 65% by widening matched inventory and ranking contracted properties first across 2–3 million daily requests	
Technical Architect	May '15 - Nov '15
CURA Software Solutions	Hyderabad
- Introduced Linux/Unix deployment for the Governance, Risk & Compliance platform, cutting running costs and winning 12 new customers within 2 months - Ported the data model from Microsoft SQL Server to MySQL, enabling the low-cost Linux deployment	
Technical Lead	Jul '13 - May '15
innRoad India Hotel Software Pvt. Ltd.	Hyderabad
- Converted a legacy ASP.NET 2.0 application into a modern single-page application using Knockout.js and a Boilerplate JavaScript framework - Productized REST-based WCF services that enabled and monetised customer integrations - Codified engineering standards and mentored the team through a full product-modernisation effort	
Software Architect	Jun '12 - May '13
McDonald's Corporation	OakBrook, IL
- Modeled the schema and governance framework for an enterprise Master Data Management (MDM) program - Forged interactive user-interface screens with drill-down, dynamic drop-downs, and full row-level editing - Partnered with data owners to deliver a robust enterprise data-governance system across the organisation	
Senior Java Developer	Apr '08 - May '12
El Paso Corporation	Houston, TX

- Rebuilt a legacy Citrix-hosted Java application into a modern **.NET 2.0** stack on **SQL Server 2005**
- Steered a scalable, maintainable .NET platform that replaced the outdated system while preserving core business functions
- Anchored large-scale enterprise modernisation for a high-stakes energy-sector client

.NET Developer

Apr '07 - Apr '08

LyntonWeb

Houston, TX

- Developed Paymaster, a **PayPal**-style payments application, for a Mexican financial institution

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning

Jan '21

IIIT Bangalore

Master's in Computer Science

Aug '04 - Dec '07

University of Houston

Houston, Texas

CERTIFICATION

PG Diploma in Artificial Intelligence and Machine Learning, IIIT Bangalore